linear inch of the screen through which the grains pass when they are graded. There are four different groups of the grain size namely coarse, medium, fine and very fine. If the grit number is large, the size of the abrasive is fine and a small grit number indicates a large grain of abrasive.

Grain size	Grain numbers
Coarse	10 12 14 16 24
Medium	30 36 46 54
Fine	80 100 120 150
Very fine	220 240 280 320 500 600

Grade

The grade of grinding wheel refers to the hardness with which the wheel holds the abrasive grains in place. It does not refer to the hardness of the abrasive grains. The grade is indicated by a letter of the English alphabet. The term 'soft' or 'hard' refers to the resistance a bond offers to disruption of the abrasives. A wheel from which the abrasive grains can easily be dislodged is called soft whereas the one, which holds the grains more securely, is called hard. The grade of the bond can be classified in three categories.

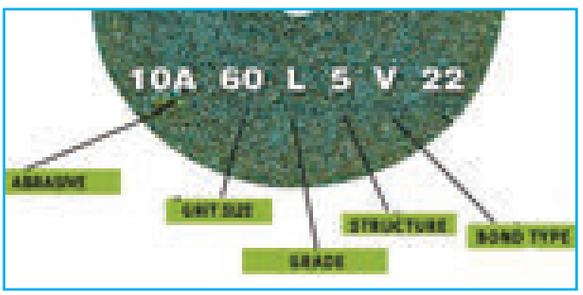
Grade	Denoted English letters
Soft	A B C D E F G H
Medium	I J K L M N O P
Hard	Q R S T U V W X Y Z

Structure

The relative spacing occupied by the abrasives and the bond is referred to as structure. It is denoted by the number and size of void spaces between grains. It may be 'dense' or 'open'. Open structured wheels are used to grinding soft and ductile materials. Dense wheels are useful in grinding brittle materials.

Structure	Identification number
Dense	1 2 3 4 5 6 7 8
Open	9 10 11 12 13 14 15 and above

4.11 Grinding wheel specification



The Indian standard marking system of grinding wheels has been prepared with a view of establishing a uniform system of marking of grinding wheels to designate their various characteristics.

Prefix	Manufacturer's abrasive type symbol
First letter	Types of abrasive
First Number	Size of abrasive
Second letter	Grade of bond
Second number	Structure of the grinding wheel
Third letter	Types of bond

For specification of grinding wheel, outside diameter, width and centre hold diameter of the grinding wheel are also to be added. The meaning of the given marking on a grinding wheel

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